

SPDM Tool Project Annual Review 2026

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Intro – SPDMM-RS

- A pure rust version SPDMM library
 - <https://github.com/ccs-spdmm-tools/spdmm-rs>
 - Following DMTF SPDMM 1.2/1.3/1.4 specification ([DSP0274](#), [DSP0277](#))
 - (SPDMM = Security Protocol and Data Model)
- with PCIe IDE_KM and TDISP feature
 - Following PCI-SIG PCIe specification (PCIe [6.2](#))
 - (IDE KM = Integrity and Data Encryption Key Management Protocol)
 - (TDISP = TEE Device Interface Security Protocol)
- Support SPDMM Requester and SPDMM Responder.
- TEE-IO Use Case

Update since last review 2025

- Add SPDM 1.4 support
 - https://www.dmtf.org/sites/default/files/standards/documents/DSP0274_1.4.0.pdf
 - All messages are upgraded to support 1.4.
 - New 1.4 capability: LARGE_RESP_CAP
 - New 1.4 PQC algorithm: ML-DSA/ML-KEM
 - Cross test with <https://github.com/DMTF/libspdm> for 1.4.
- Add PQC crypto binding – <https://github.com/aws/aws-lc-rs> for `std`
 - `no-std` support is still WIP.
- Change webpki to rustls-webpki
 - Considering removing webpki at all, just implement the X509 parser.

Plan

- FIPS 140-3
 - Add SPDM-Key-Exchange test.
 - NOTE: SPDM key exchange is approved by FISP.
 - <https://csrc.nist.gov/csrc/media/Projects/cryptographic-module-validation-program/documents/fips%20140-3/FIPS%20140-3%20IG.pdf>
- Formal Verification
 - Current rudra & mirai are not maintained and not working (on new rust toolchain).
 - Evaluating new formal tools:
 - **Kani** - <https://model-checking.github.io/kani/>
 - Bounded Model Checker - formal verification
 - **Verus** - <https://verus-lang.github.io/verus/guide/>
 - Deductive Verifier - Satisfiability Modulo Theories (STM)-based
- More Security test
 - 3rd party penetration test (Budget Request)

Adoption

- Intel TDX Connect TSM
 - Secure Session Establishment for IDE Key and TDISP message
- Intel Migration-TD (<https://github.com/intel/migtd>) for Live Migration
 - Secure Session Establishment for Migration Key

Annual Review

- Charter – No Change
- Project Status – On Track
- Budget – **May request budget for security audit (penetration test).**
- License – No Change
- [OpenSSF Best Practices Badge](#) – 97%.

Backup

Intro – Background

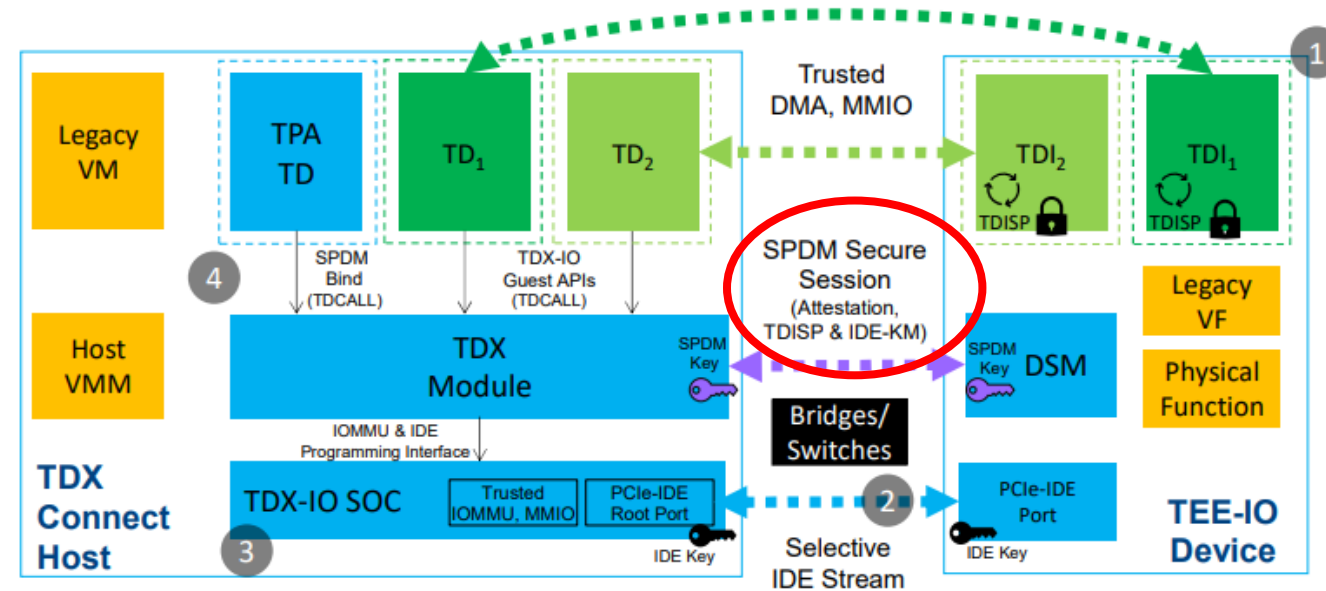
- Industry trend:
 - Move TEE-TCB from {SOC} only to {SOC + Device}.
- Change and Solution:
 - Host TEE needs to trust and verify the device. (via SPDMM protocol)
 - Host SOC and device need a secure way to encrypt PCIe TLP. (via PCIe IDE)
 - Host TEE needs a way to manage the device. (via PCIe TDISP)
- Reference:
 - [Making PCI devices ready for confidential computing](#), OC3, 2023
 - PCIe Base Specification [6.2](#), January, 2024

Intro – SPDM-RS

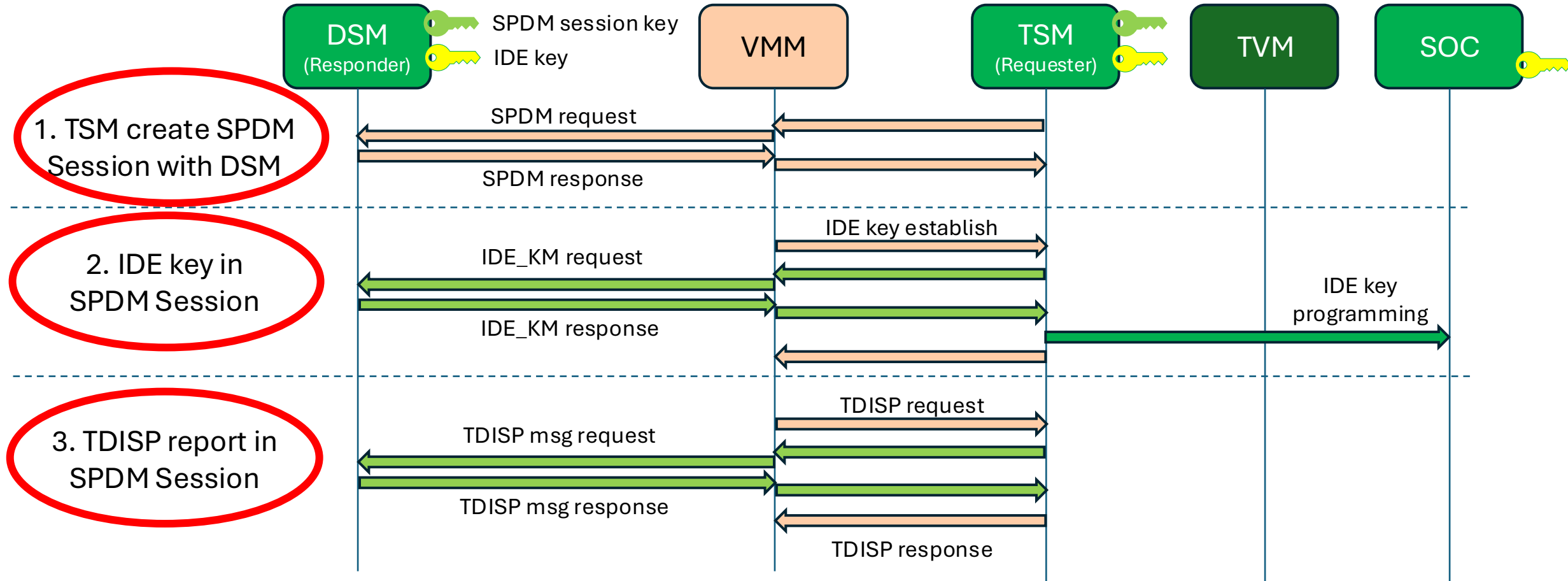
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 - <https://github.com/ccs-spdm-tools/spdm-rs>
 - Following DMTF SPDM 1.2 specification ([DSP0274](#), [DSP0277](#))
 - (SPDM = Security Protocol and Data Model)
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- TEE-IO Use Case (see next pages)

SPDM stack for TSM

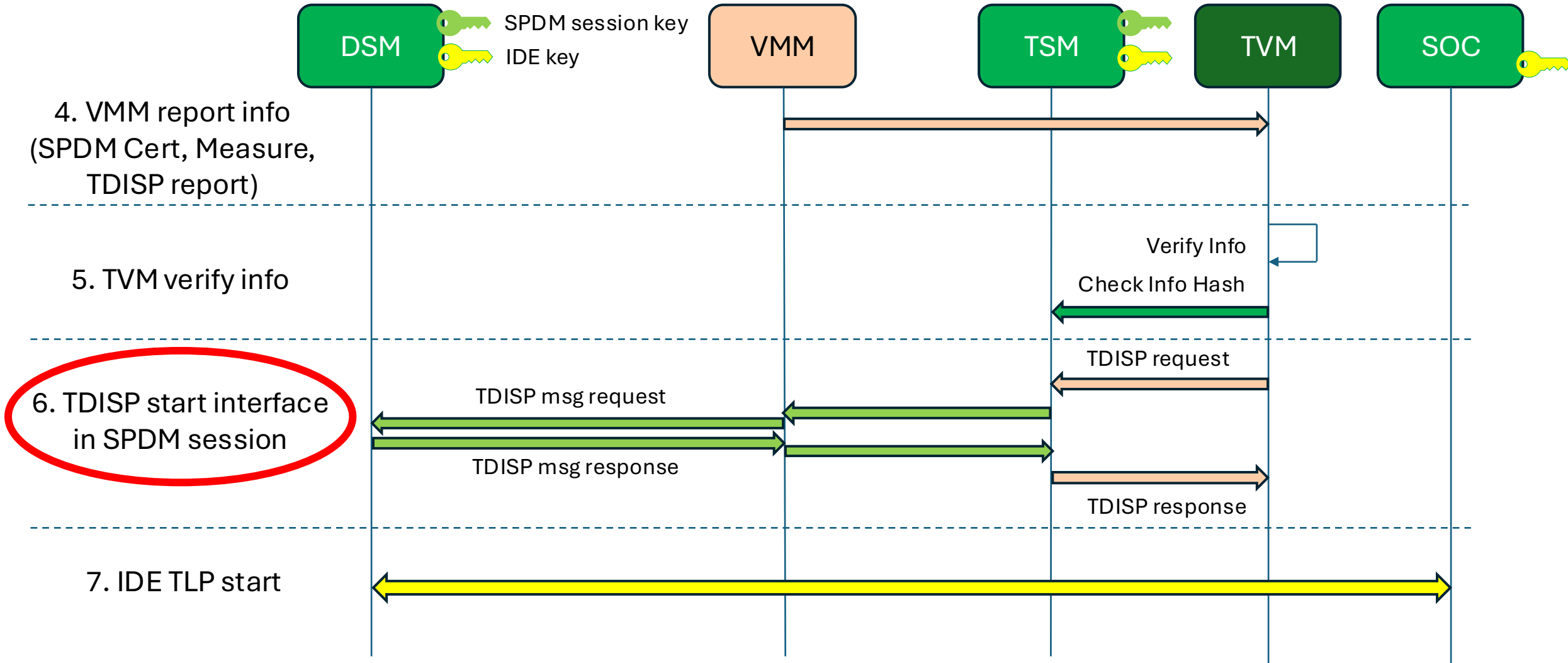
- Intel TSM (TDX-module + TPA)
 - [Intel® TDX Connect Architecture Specification](#)
 - SPDM Requester
 - SPDM Device Certificate and Measurement Collection
 - SPDM Session with the device, for IDE_KM and TDISP
- SPDM candidate for any other secure TSM.



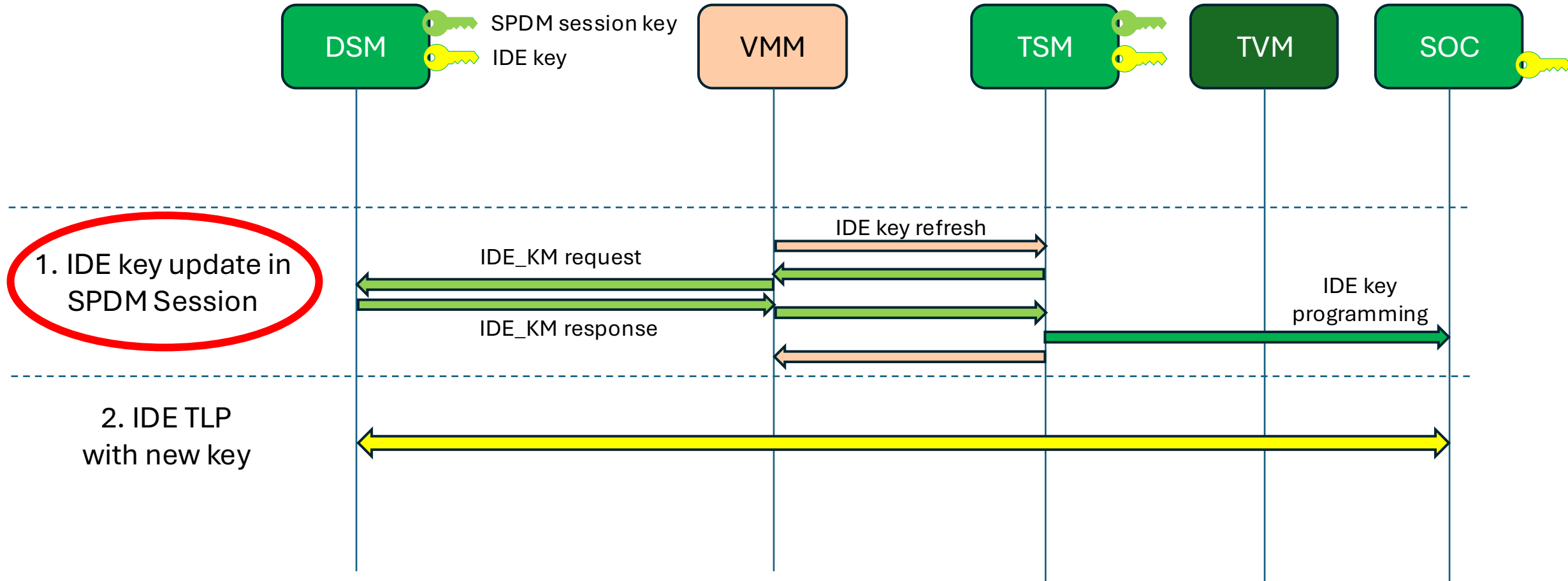
TEE-IO Use Case - Startup



TEE-IO Use Case - Startup (Cont'd)



TEE-IO Use Case - IDE Key Refresh



Update since adoption

- Upgrading to SPDMM 1.3
 - VCA, DIGEST, MEASUREMENT, CHALLENGE, KEY_EXCHANGE, PSK_EXCHANGE
- Adding FIPS self-test.
 - HASH, AEAD (AES-GCM), RSA, ECDSA
- Bug fix
 - Fix bugs in certificate chain verification, key update, etc.
 - Add more unit test.
 - Add more fuzzing test.

Plan

- SPDM version upgrade
 - Add new SPDM 1.3 feature
 - Prepare for SPDM 1.4 – PQC
- FIPS 140-3
 - Add rest FIPS crypto self-test
 - Add SPDM-Key-Exchange self-test
- More Security test
 - 3rd party penetration test (Budget Request)

Adoption in planning

- Intel TDX Connect TSM
- Rivos Salus TSM

Annual Review

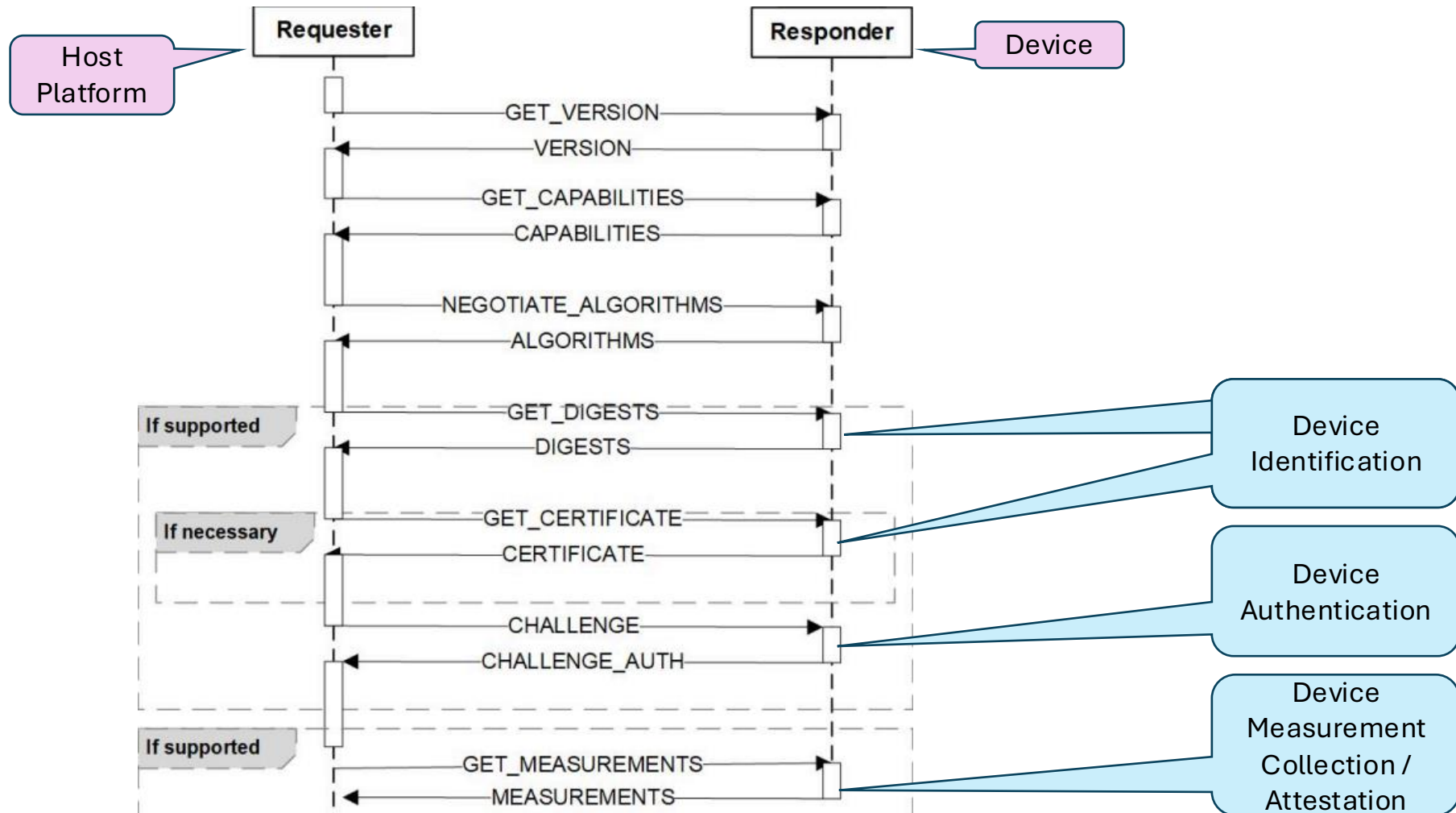
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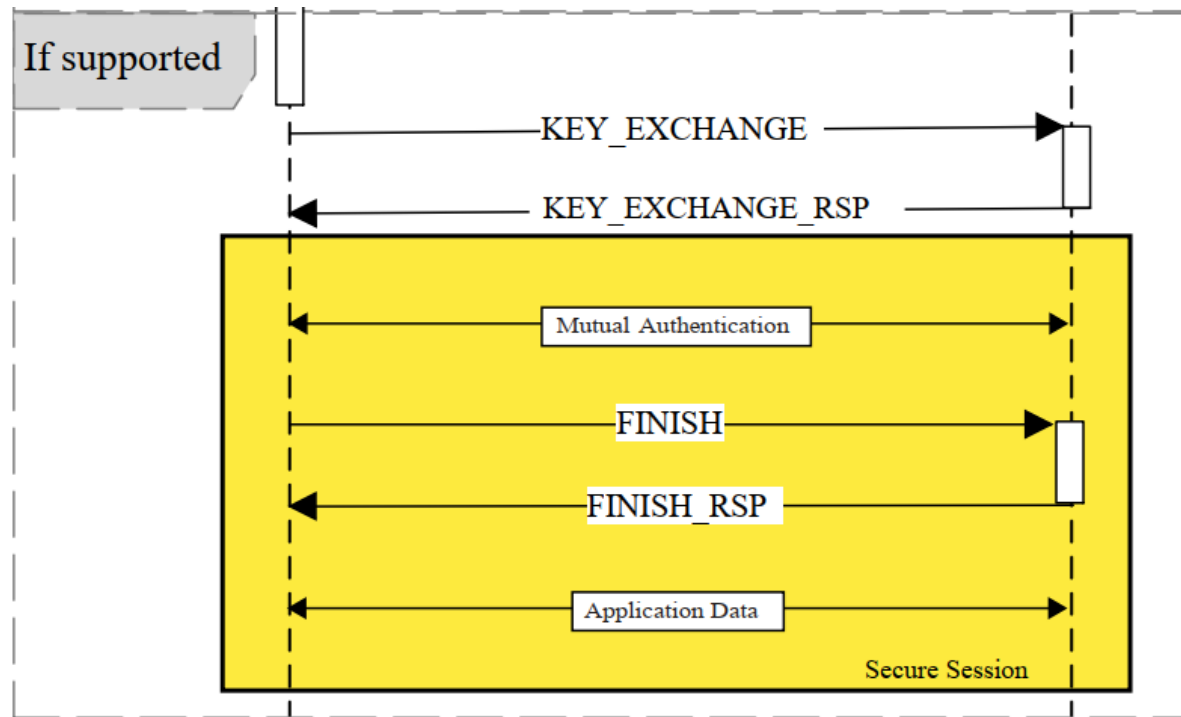
What is SPDm?

- Security Protocol and Data Model (SPDM)
 - DSP0274 - <https://www.dmtf.org/dsp/DSP0274>
 - Defined by DMTF SPDM WG - <https://www.dmtf.org/standards/spdm>
- Version
 - SPDM 1.0 (Dec 2019) – Device Authentication and Measurement.
 - SPDM 1.1 (August 2020) – Device Secure Session Communication.
 - SPDM 1.2 (Dec 2021) – DICE support, Device Provisioning, Message Chunking.
 - SPDM 1.3 (May 2023) – Multiple Key support, Hash Extend Measurement (HEM), Measurement Extension Log (MEL), Async Event.
- Adoption
 - DMTF, PCI/CXL, TCG, MIPI, NVMe, ...

SPDM 1.0



SPDM 1.1



Diffie-Hellman (DH)
based Key Exchange +
(Mutual) Authentication

or

Pre-Shared Key (PSK)
based Key Exchange

SPDM use case in Confidential Computing

- A. Device Attestation
 - PCI Component Measurement and Authentication (CMA)
- B. Device Secure Communication
 - PCI Integrity and Data Encryption (IDE) Key Management (KM) – IDE_KM
 - PCI TEE Device Interface Security Protocol (TDISP)
- C. Secure Communication between two TEEs
 - Candidate: TLS, SPDM, ...

Rust-SPDM Overview

- The Rust implementation of SPDM protocol.
 - <https://github.com/intel/rust-spdm>
 - Apache 2.0 License
 - SPDM version 1.2
 - Support SPDM Requester and SPDM Responder
 - Support rust no-std.
 - Use [ring/webpki](#) crypto stub by default.
 - [rust-mbedtls](#) as backup.
 - Interoperability test with DMTF [libspdm](#) sample implementation.